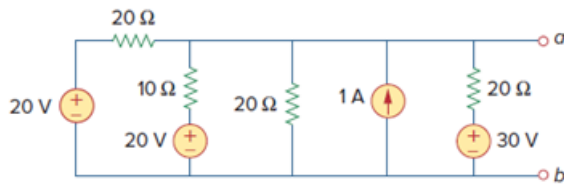


Problem

Use source transformation to reduce the circuit between terminals a and b shown in Fig. 4.88 to a single voltage source in series with a single resistor.

Figure. 4.88:

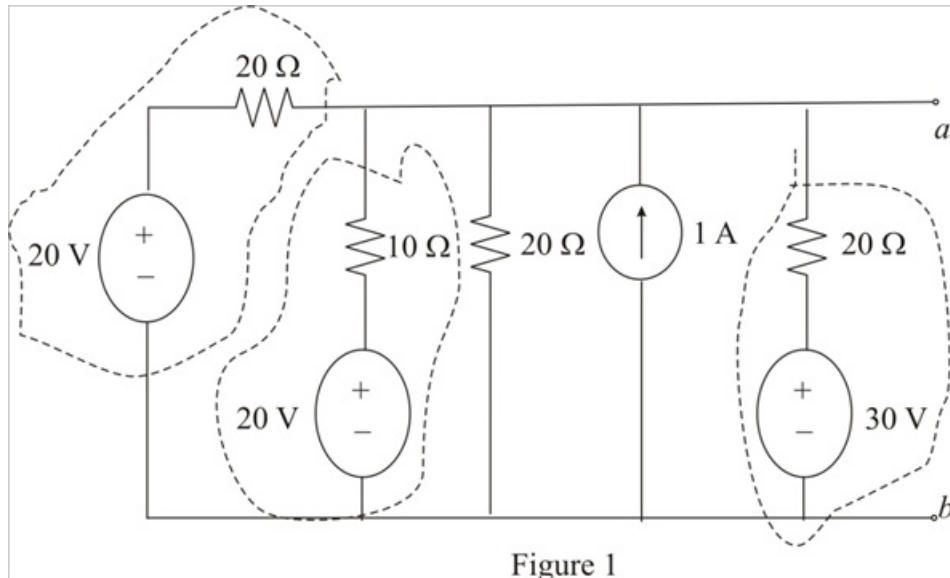


Step-by-step solution

Step 1 of 10

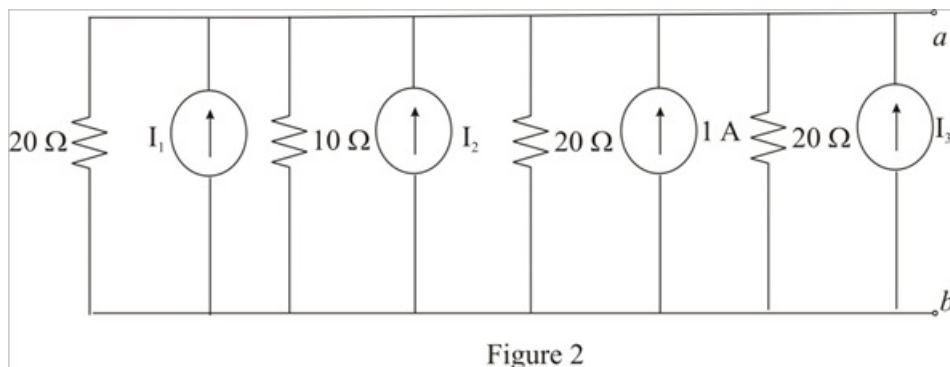
Refer to Figure 4.88 in the text book.

Consider the circuit diagram shown in Figure 1.



Step 2 of 10

Convert all the voltage sources to current sources using source transformation technique as shown in Figure 2.



Step 3 of 10

Calculate the value of the current, I_1 .

$$\begin{aligned} I_1 &= \frac{20}{20} \\ &= 1 \text{ A} \end{aligned}$$

Therefore, the value of the current I_1 is 1 A.

Step 4 of 10

Calculate the value of the current, I_2 .

$$\begin{aligned} I_2 &= \frac{20}{10} \\ &= 2 \text{ A} \end{aligned}$$

Therefore, the value of the current I_2 is 2 A.

Step 5 of 10

Calculate the value of the current, I_3 .

$$\begin{aligned} I_3 &= \frac{30}{20} \\ &= 1.5 \text{ A} \end{aligned}$$

Therefore, the value of the current I_3 is 3 A.

Step 6 of 10

Represent the circuit diagram as a single current source in parallel to a resistance as shown in Figure 3.

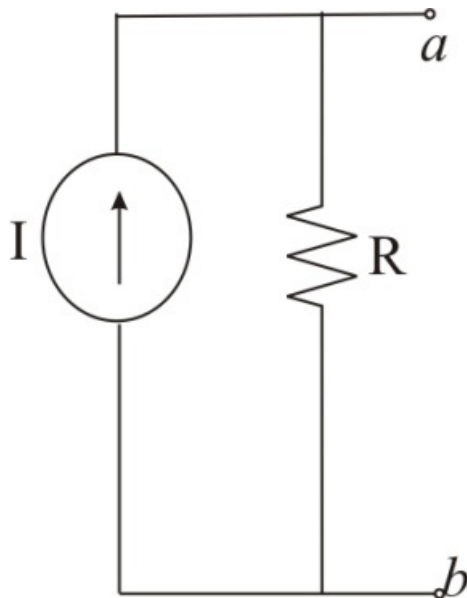


Figure 3

Step 7 of 10

Calculate the value of current, I_3 .

$$I = I_1 + I_2 + I_3 + 1$$

Substitute 1.5 A for I_3 , 2 A for I_2 and 1 A for I_1 in the equation.

$$\begin{aligned} I &= 1 + 2 + 1.5 + 1 \\ &= 5.5 \text{ A} \end{aligned}$$

Therefore, the value of the current I is 5.5 A .

Step 8 of 10

Calculate the value of the resistance, R .

$$\begin{aligned} R &= 20 \parallel 10 \parallel 20 \parallel 20 \\ &= \frac{1}{\frac{1}{20} + \frac{1}{10} + \frac{1}{20} + \frac{1}{20}} \\ &= \frac{1}{0.05 + 0.1 + 0.05 + 0.05} \\ &= \frac{1}{0.25} \end{aligned}$$

$$R = 4 \Omega$$

Therefore, the value of the resistance R is 4Ω .

Step 9 of 10

Convert the current source to voltage source using source transformation.

$$V = IR$$

Substitute 4Ω for R and 5.5 A for I in the equation.

$$\begin{aligned} V &= (5.5)(4) \\ &= 22 \text{ V} \end{aligned}$$

Therefore, the value of the voltage source V is 22 V .

Step 10 of 10

Represent the circuit diagram as a single voltage source in series with a resistance as shown in Figure 4.

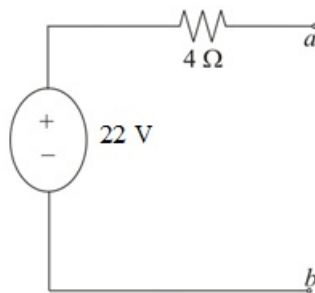


Figure 4

Therefore, the required circuit is obtained using source transformation technique and is shown in Figure 4.